

Data Flow Diagram

Date	10 July 2026
Team ID	xxxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Data Flow Diagram — Description (see legend below; a visual diagram image should accompany this text in the final submission).

Symbol	Name	Description
Oval	External Entity	Cardholder, Payment Gateway, Compliance Officer
Rectangle w/ number	Process	1. Capture Transaction, 2. Preprocess Features, 3. Score with ML Model, 4. Flag/Approve, 5. Notify Analyst
Solid rectangle	Data Store	Transaction Log DB, Trained Model Store (.pkl), Alert/Case DB
Labeled arrow	Data Flow	Raw transaction → preprocessed features → fraud probability → decision → alert/report

Flow narrative: (1) Payment Gateway sends raw transaction (Time, Amount, V1–V28) to the Preprocessing process. (2) Preprocessing scales Time/Amount and forwards the feature vector to the ML Scoring process, which loads the trained model from the Model Store. (3) The Scoring process outputs a fraud probability to the Decision process, which either auto-approves or raises a flag. (4) Flagged transactions are written to the Alert/Case DB and pushed to the Analyst Dashboard for the Compliance Officer to review. (5) All transactions (flagged or not) are logged to the Transaction Log DB for audit and future retraining.